

TsaoSciResearcher Engineering Audit

Release 0.7.4 | preflight evidence | 2026-09-11

Evidence-first scientific research control layer

Question -> evidence -> design -> execution handoff -> validation -> artifact

Truth boundary: a contract or handoff is not an execution receipt.

341

Capability contracts

322

Named catalog contracts

15

Workflows

20

Schemas

52

Test modules

7

Domain packs

Delivered integration

1. Measurement Boundary with calibration, uncertainty and applicability fields
2. Structure-Property planning with confounders, alternatives and scale bridges
3. Bilingual causality guard and evidence-to-source execution-receipt checks
4. Deterministic test/quality HTML-SVG interfaces and this four-page PDF

Repository operation

All changes were written directly to main; no branch and no pull request were created.

Scoped software validation matrix

Recorded cross-platform baseline

These platform results are a recorded baseline, not a fresh current-tree run.

macos python 3 12		PASS
ubuntu python 3 10		PASS
ubuntu python 3 13		PASS
windows python 3 12		PASS

Scoped validation gates

bandit high severity		NOT RUN
bounded performance		LOCAL_PREFLIGHT
byte identical release builds		LOCAL_PREFLIGHT
complete regression		PARTIAL
coverage line and branch		LOCAL_PREFLIGHT
critical mutation killed		NOT RUN
dependency vulnerability audit		NOT RUN
deterministic sbom		PASS
deterministic visual reports		PASS
docs build		NOT RUN
execution receipts		LOCAL_PREFLIGHT
json schemas 19		PASS
mypy strict		NOT RUN
repository and contract audit		PASS
reproducibility capsule		LOCAL_PREFLIGHT
reverse order regression		NOT RUN
ruff format and lint		NOT RUN
scientific quality guards		PASS
seeded random order regression		NOT RUN
validation tree digest		PASS
wheel and sdist install		NOT RUN

Scoped status applies only to the evidence levels explicitly recorded above.

Scientific quality controls

Four deterministic contracts: measurement, multiscale reasoning, causal claims and evidence provenance.

Examples: PASS 3 | WARN 0 | BLOCK 1 | average completeness 89%

Measurement boundary

PASS

Declared completeness: 100%

Operational definition, calibration, uncertainty and applicability are explicit.

MB-COMPLETE: Operational measurement boundary is explicit.

Kind: measurement-boundary

Structure-property planner

PASS

Declared completeness: 100%

The chain records intervention, structure, mediator, response and competing...

SP-CHAIN: Structure, mediator and property are all explicit.

Kind: structure-property-plan

Causality guard

BLOCK

Declared completeness: 57%

Association-only evidence blocks causal wording in English and Chinese.

CG-OVERCLAIM: Causal wording exceeds the declared study design.

Kind: causality-guard

Evidence traceability

PASS

Declared completeness: 100%

Claims link to evidence locators and execution claims require receipts.

ET-COMPLETE: Claim, evidence, locators and execution boundary are traceable.

Kind: evidence-traceability

BLOCK is intentional when wording exceeds evidence or an execution claim lacks a receipt.

The guards remain configurable and do not hard-code material-specific trends.

Scoped evidence and limitations

Recorded validation scope

Scope: preflight

Only the explicitly recorded checks may be interpreted as passing.

Current-tree end-to-end CI

Status: not recorded

No exact commit attestation is asserted until external CI artifacts record it.

Explicit limitations

1. Checked-in preflight evidence is not a substitute for the external CI attestation.
2. External scientific execution requires a checksum-verifiable execution receipt.

Reproduction entry points

```
python scripts/build_validation_evidence.py --check
```

```
python scripts/build_test_dashboard.py --check
```

```
python scripts/build_research_quality_dashboard.py --check
```

```
python scripts/build_engineering_report.py --check
```

Software validation is not acceptance of experiments, external calculations, legal or safety claims.